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- <https://nvemangue.github.io/ru/>



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$$\left\{ \begin{array}{l} \frac{dx}{dt} = -0.29x(t) + 0.039x(t)y(t) \\ \frac{dy}{dt} = 0.49y(t) - 0.059x(t)y(t) \end{array} \right.$$

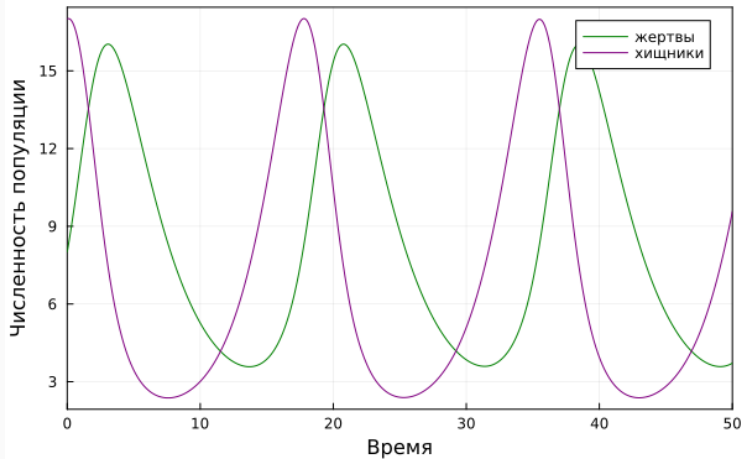
,
: $x_0 = 8, y_0 = 17$.



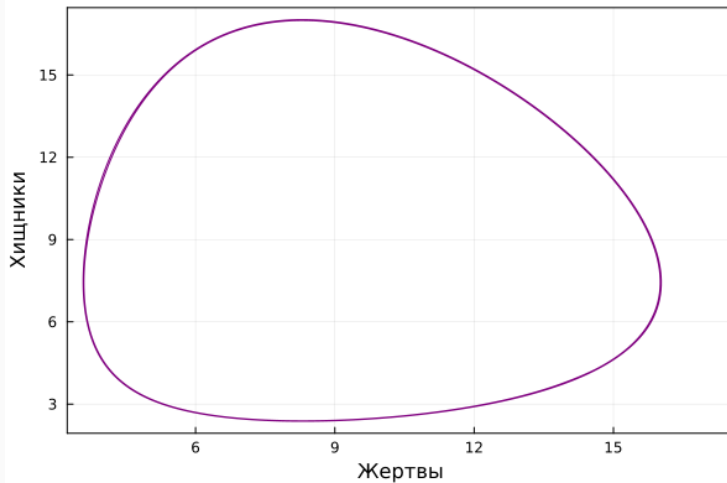
```
#
using DifferentialEquations, Plots;
#
function LV(u, p, t)
    x, y = u
    a, b, c, d = p
    dx = a*x - b*x*y
    dy = -c*y + d*x*y
    return [dx, dy]
end
```

```
#  
u0 = [8,17]  
p = [-0.29, -0.039, -0.49, -0.059]  
tspan = (0.0, 50.0)  
prob = ODEProblem(LV, u0, tspan, p)  
sol = solve(prob, Tsit5())
```


Модель Лотки-Вольтерры

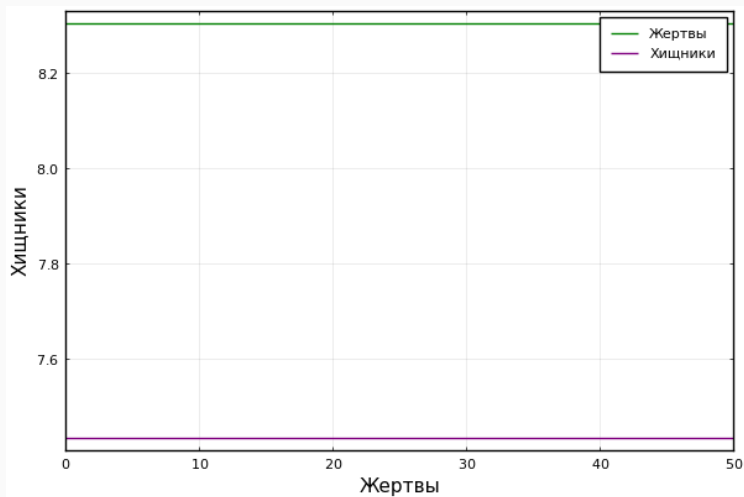


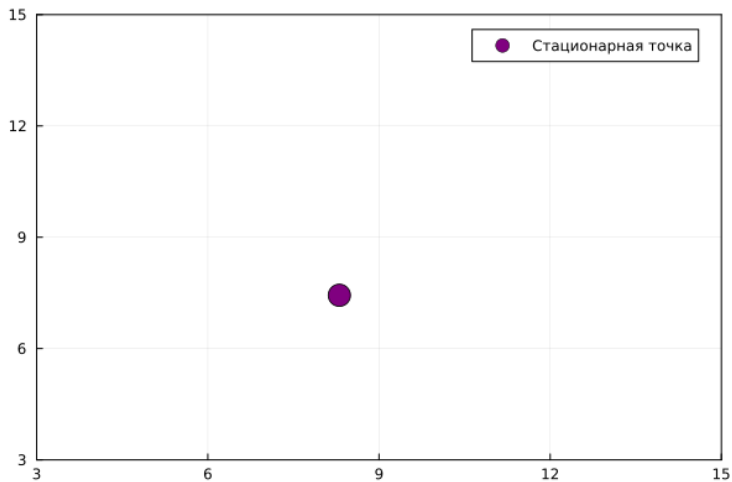
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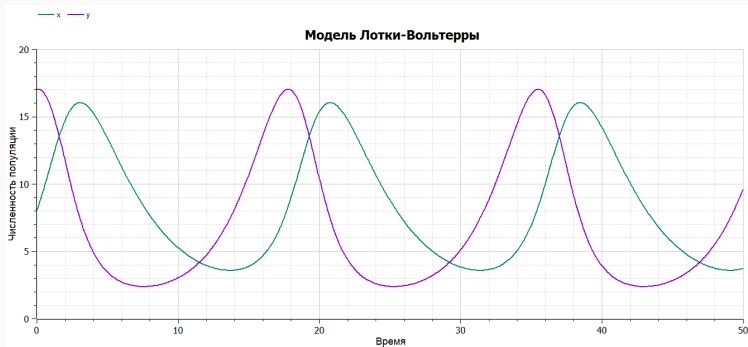
$$\left\{ \begin{array}{l} x_0 = \frac{\gamma}{\delta} \\ y_0 = \frac{\alpha}{\beta} \end{array} \right.$$

$$, \quad x_0 = \frac{0.29}{0.039} = 8.305084745762713, \quad y_0 = \frac{0.49}{0.059} = 7.435897435897435$$



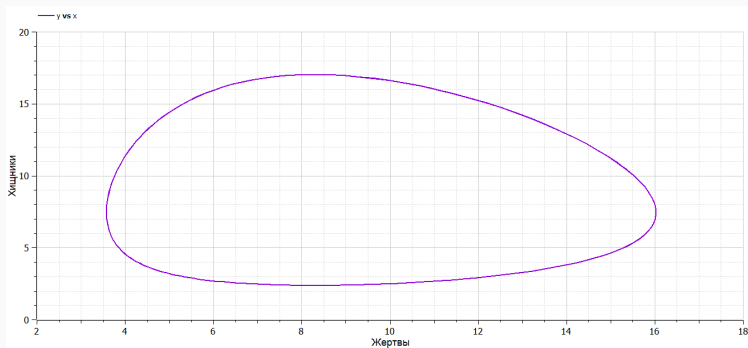


```
model lab5_1
  parameter Real a = -0.29;
  parameter Real b = -0.039;
  parameter Real c = -0.49;
  parameter Real d = -0.059;
  parameter Real x0 = 8;
  parameter Real y0 = 17;
  Real x(start=x0);
  Real y(start=y0);
equation
  der(x) = a*x - b*x*y;
  der(y) = -c*y + d*x*y;
end lab5_1;
```



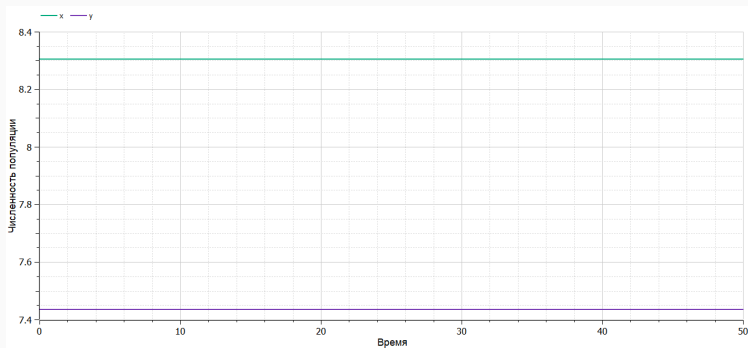
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https://ru.wikipedia.org/wiki/—_—_ .